A Theory of Economic Slack

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APPENDIX I.

Inflation and tightness dynamics at the zero lower bound

In this appendix, we derive and study the properties of the dynamical system describing the solution to the slackish business cycle model of chapter 14 at the zero lower bound. In normal times, the central bank sets a real interest rate, which separates the dynamics of inflation and tightness. Tightness can therefore be determined as in the fixed-inflation model of chapter 13, and inflation can be determined from the Phillips curve. At the zero lower bound, this separation disappears as the central bank loses control of the real interest rate. The nominal interest rate is stuck at 0 so the real interest rate is just negative inflation: $r = i - \pi = -\pi$. Tightness and inflation are therefore jointly determined by a two-dimensional dynamical system.

I.1. Relationships between inflation, tightness, and the costate variables

The first step in building the two-dimensional dynamical system describing the model of chapter 14 at the zero lower bound is to express the costate variables ψ^b and ψ^p as a function of inflation π and tightness θ. Using these relationships, we will be able to transform the differential equations for the costate variables into differential equations for inflation and tightness.

This is easy to do for the costate variable ψ^p, as the relationship is directly given by equation (14.16):

$$\psi^p = \varpi \left[\pi - \pi^*\right] \tag{I.1}$$

$$\dot{\psi}^p = \varpi\dot{\pi}. \tag{I.2}$$

For the costate variable ψ^b, the relationship arises from equation (14.15). Indeed, in the model, aggregate consumption is directly determined by tightness:

$$c(\theta) = \frac{y(\theta)}{1+\tau(\theta)} = \frac{1-u(\theta)}{1+\tau(\theta)} \cdot kh. \tag{I.3}$$

Using the function $c(\theta)$, we can relate ψ^b to tightness using (14.15):

$$\psi^b = (1-\alpha) \cdot \frac{c(\theta)^{-\alpha}}{1+\tau(\theta)}. \tag{I.4}$$

Notice also, which will be useful later, that the product $\psi^b(\theta)y(\theta)$ is simply given by:

$$\psi^b(\theta)y(\theta) = \psi^b(\theta)c(\theta)[1+\tau(\theta)] = (1-\alpha)c(\theta)^{1-\alpha}. \tag{I.5}$$

Using a basic chain rule, the time derivative of the costate variable therefore is:

$$\dot{\psi}^b = \frac{\psi^b}{\theta} \cdot \epsilon_\theta^{\psi^b} \cdot \dot{\theta},$$

where $\epsilon_\theta^{\psi^b}$ is the elasticity of the costate ψ^b with respect to tightness θ.

To compute that elasticity (and others below), we use the result that the elasticity of $1+\tau(\theta)$ with respect to θ is given by (6.12):

$$\epsilon_\theta^{1+\tau} = \eta(\theta)\tau(\theta). \tag{I.6}$$

Another useful elasticity is the elasticity of $1-u(\theta)$ with respect to θ, which can be inferred from (13.13):

$$\epsilon_\theta^{1-u} = \frac{-u(\theta)}{1-u(\theta)} \cdot \epsilon_\theta^u = [1-\eta(\theta)]\,u(\theta). \tag{I.7}$$

From these results and (I.3), we infer that the elasticity of $c(\theta)$ with respect to θ is

$$\epsilon_\theta^c = [1-\eta(\theta)]\,u(\theta) - \eta(\theta)\tau(\theta).$$

Then, from these results and (I.4), we infer that the elasticity of ψ^b with respect to θ is

$$\epsilon_\theta^{\psi^b} = -\left[\alpha\left[1-\eta(\theta)\right]u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right].$$

And finally from this we relate the time derivative of the costate variable ψ^b to that of tightness θ:

$$\frac{\dot{\psi}^b}{\psi^b} = -\left[\alpha\left[1-\eta(\theta)\right]u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right]\cdot\frac{\dot{\theta}}{\theta}. \tag{I.8}$$

I.2. Differential equation for inflation

Using the results from the previous section, we now transform the differential equation for the costate variable ψ^p, given by (14.18), into a differential equation for inflation π. We simply replace ψ^p and $\dot{\psi}^p$ in the equation by the expressions (I.1) and (I.2), and $\psi^b y$ by the expression (I.5). We obtain:

$$\dot{\pi} = \delta\left[\pi-\pi^*\right] + \frac{1-\alpha}{\varpi}\cdot c(\theta)^{1-\alpha}\left[1-\frac{\eta(\theta)}{1-\eta(\theta)}\cdot\frac{\tau(\theta)}{u(\theta)}\right]. \tag{I.9}$$

Accordingly, the nullcline $\dot{\pi}=0$ is given by:

$$\pi = \pi^* + \frac{1-\alpha}{\delta\varpi}\cdot c(\theta)^{1-\alpha}\left[\frac{\eta(\theta)}{1-\eta(\theta)}\cdot\frac{\tau(\theta)}{u(\theta)}-1\right]. \tag{I.10}$$

This expression is close but not the same as the Phillips curve in normal times, given by equation (14.20). This is because the costate variable ψ^b is fixed, determined by the real rate in normal times, whereas it is jointly determined by tightness and inflation at the zero lower bound.

I.3. Differential equation for tightness

Next, we transform the differential equation for the costate variable ψ^b, given by (14.17), into a differential equation for tightness θ. First, we divide the equation by ψ^b. We also set the real interest rate to $r=-\pi$. We obtain

$$\frac{\dot{\psi}^b}{\psi^b} = (\delta+\pi) - \frac{1}{a\psi^b}.$$

Next, we replace ψ^b and $\dot{\psi}^b$ in the equation by the expressions (I.4) and (I.8). This yields:

$$\left[\alpha\left[1-\eta(\theta)\right]u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)\right]\cdot\frac{\dot{\theta}}{\theta} = \frac{1}{a(1-\alpha)}\cdot\left[1+\tau(\theta)\right]c(\theta)^{\alpha} - (\delta+\pi). \tag{I.11}$$

From this equation, we see that the nullcline $\dot{\theta} = 0$ is given by:

$$\pi = \frac{1}{a(1-\alpha)} \cdot [1 + \tau(\theta)]\, c(\theta)^{\alpha} - \delta. \tag{I.12}$$

I.4. Equilibrium of the dynamical system

We now determine the equilibrium of the dynamical system composed of differential equations (I.9) and (I.11). At the equilibrium, $\dot{\pi} = \dot{\theta} = 0$, so the equilibrium is determined by equations (I.10) and (I.12).

Let us introduce a few auxiliary functions to simplify the analysis:

$$P(\theta) = \pi^* + \frac{1-\alpha}{\delta\varpi} \cdot c(\theta)^{1-\alpha}\left[\frac{\eta(\theta)}{1-\eta(\theta)} \cdot \frac{\tau(\theta)}{u(\theta)} - 1\right]$$
$$Q(\theta) = \frac{1}{a(1-\alpha)} \cdot [1 + \tau(\theta)]\, c(\theta)^{\alpha} - \delta$$
$$R(\theta) = \frac{\theta}{\alpha\,[1-\eta(\theta)]\, u(\theta) + (1-\alpha)\eta(\theta)\tau(\theta)}.$$

With these auxiliary functions, the nullcline (I.10) is given by $\pi = P(\theta)$ and the nullcline (I.12) is given by $\pi = Q(\theta)$. Let $S(\theta) = Q(\theta) - P(\theta)$. The tightness that solves the model is therefore the solution to $S(\theta) = 0$.

At $\theta = 0$, we have $u(0) = 1$ so $c(0) = 0$ and $S(0) = -(\delta + \pi^*) < 0$.

At $\theta = \theta^*$, we have $\tau/u = (1-\eta)/\eta$, from (9.11), so that

$$S(\theta^*) = \frac{1}{a(1-\alpha)} \cdot \left[1 + \tau(\theta^*)\right] c(\theta^*)^{\alpha} - (\delta + \pi^*).$$

Because we are at the zero lower bound, we know that the marginal utility of wealth $1/a$ is quite large—otherwise a positive nominal interest rate would be sufficient to maintain the economy at the divine equilibrium, where $\theta = \theta^*$ and $\pi = \pi^*$. Studying the zero lower bound only makes sense when aggregate demand is depressed enough because of a high marginal utility of wealth. Hence, the economy enters the zero lower bound only when at a zero nominal interest rate, aggregate demand is below the efficient level of output: $y^d(\theta^*, -\pi^*) < y(\theta^*)$. Rearranging this inequality imposes that the marginal utility of wealth satisfies:

$$\frac{1}{a} > \frac{(\delta + \pi^*)(1-\alpha)c(\theta^*)^{-\alpha}}{[1 + \tau(\theta^*)]}.$$

Hence, when the marginal utility of wealth is high enough to push the economy at the zero lower bound, than it is high enough so that $S(\theta^*) > 0$.

We have established that $S(0) < 0$ and $S(\theta^*) > 0$ and the function S is clearly continuous on $[0, \theta^*]$. According to Bolzano's theorem, the equation $S(\theta) = 0$ therefore admits a solution on $(0, \theta^*)$ (result A.1). This means that the dynamical system admits an equilibrium on $(0, \theta^*)$, which we denote (θ^z, π^z). Since $S(0) < 0$, we know that when S crosses 0 from below at the equilibrium, so $S'(\theta^z) > 0$.

Is the equilibrium unique? It would be if S was strictly increasing on $(0, \theta^*)$. The function Q is strictly increasing on $(0, \theta^*)$, as $\tau(\theta)$ and $c(\theta)$ are increasing on the interval. But the function P is not monotonous (recall that $P(0) = P(\theta^*) = \pi^*$), so we cannot say for sure that $S(\theta) = Q(\theta) - P(\theta)$ is strictly increasing over the entire interval.

However, when prices are rigid enough, so ϖ is large enough, the fluctuations in P can be made arbitrarily small. Indeed, at the limit where $\varpi \to \infty$, $P(\theta) \to \pi^*$ for any θ. So for ϖ large enough, the fluctuations in P can be made small enough that S never returns to 0 once it has crossed 0 at θ^z—in which case the equilibrium of the system is unique. The equilibrium is for instance unique under the calibrations presented in figure 14.2. In the rest of the appendix we assume that price rigidity ϖ is sufficient to guarantee that the equilibrium is unique.

I.5. Classification of the dynamical system

In this section, we determine whether, around the zero-lower-bound equilibrium, the dynamical system is a source, sink, or saddle. To do that, we need to compute the Jacobian of the system (appendix A.8.3), which describes the local dynamics around the zero-lower-bound equilibrium.

With the above auxiliary functions, the differential equation (I.11) is given by

$$\dot{\theta} = R(\theta) \cdot [Q(\theta) - \pi]$$

and the differential equation (I.9) is given by

$$\dot{\pi} = \delta \left[\pi - P(\theta)\right].$$

With this notation the Jacobian of the system is

$$\mathbf{J} = \begin{bmatrix} \delta & -\delta P'(\theta^z) \\ -R(\theta^z) & R(\theta^z) Q'(\theta^z) \end{bmatrix}.$$

(We simplified it because $Q(\theta^z) - \pi^z = 0$ by definition of the equilibrium.)

To describe the nature of the dynamics of inflation and tightness around the zero-lower-bound equilibrium, we must compute the trace and determinant of the Jacobian.

We find:

$$\begin{aligned}
\mathrm{tr}(\mathbf{J}) &= \delta + R(\theta^z)Q'(\theta^z)\\
\det(\mathbf{J}) &= \delta R(\theta^z)\left[Q'(\theta^z) - P'(\theta^z)\right] = \delta R(\theta^z)S'(\theta^z).
\end{aligned}$$

We know that $\delta > 0$, $R(\theta^z) > 0$, and $Q'(\theta^z) > 0$, so the trace of the Jacobian is strictly positive. We have also seen that $S'(\theta^z) > 0$, so the determinant of the Jacobian is strictly positive. Accordingly, the dynamical system describing inflation and tightness is a source around the zero-lower-bound equilibrium (result A.33).